

Group Theory

Group Theory Notes

Useful Facts

- Let G be a finite group, and suppose G acts on itself by conjugation. Then, the stabilizer of $x \in G$ is denoted $C_G(x)$, and by the orbit-stabilizer theorem, it follows that the number of *elements* of the conjugacy class of x , $K_G(x)$, is given by $[G : C_G(x)]$. The *class equation* says we may find

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(x_i)],$$

where $\{x_1, \dots, x_r\}$ is a family of representatives of conjugacy classes.

- If G acts on its *subgroups* by conjugation, then the *normalizer* of $H \leq G$, denoted $N_G(H)$, is the stabilizer of H under conjugation. Similarly, number of subgroups conjugate to $H \leq G$ is given by $[G : N_G(H)]$.
- If $H \trianglelefteq G$ is a normal subgroup, then G acts on H by conjugation. The corresponding permutation $\rho: G \rightarrow \text{Aut}(H)$ has kernel $C_G(H)$, where

$$C_G(H) = \{g \in G : ghg^{-1} = h \text{ for all } h \in H\}.$$

- A subgroup $H \leq G$ is called *characteristic* if $\sigma(H) = H$ for all automorphisms $\sigma \in \text{Aut}(G)$. If $K \leq H \trianglelefteq G$ is a chain of subgroups with K characteristic in H , while H is normal in G , then $K \trianglelefteq G$ is normal.

The subgroups $[G, G]$ and $Z(G)$ are characteristic in G .

- Every p -group has nontrivial center.
- If G acts on a set X , and H is the stabilizer subgroup of some $x \in X$, then if $\rho: G \rightarrow \text{Sym}(X)$ is the permutation representation for this action, then $\ker(\rho) \leq H$.
- If $P_1, \dots, P_r$ are p -Sylow subgroups of G , they are all conjugate to each other, and the normalizer $N_G(P_i)$ is a conjugate of $N_G(P_j)$ for $1 \leq i, j \leq r$.
- Every conjugacy class in the symmetric group S_n is given by the cycle type. Moreover, given an element $\sigma \in A_n$, the conjugacy classes $K_{A_n}(\sigma) = K_{S_n}(\sigma)$ if and only if there is an odd permutation in the centralizer $C_{S_n}(\sigma)$. If there is no odd permutation in $C_{S_n}(\sigma)$, then the conjugacy class $K_{S_n}(\sigma)$ splits into two separate conjugacy classes over A_n .

Semidirect Products

Suppose N and H are subgroups of a group G , where N is normal and H is such that $N \cap H = \{e\}$. We will obtain a description of the structure of G in terms of the structure of N and H .

A short exact sequence of groups is a sequence of groups and homomorphisms

$$1 \rightarrow N \xrightarrow{\varphi} G \xrightarrow{\psi} H \rightarrow 1$$

where ψ is surjective and φ identifies N with $\ker(\psi)$. In other words, the sequence is exact if N is normal in G and ψ induces an isomorphism $G/N \cong H$.

Definition: Let N and H be groups. A group G is an *extension* of H by N if there is an exact sequence of groups

$$1 \rightarrow N \rightarrow G \rightarrow H \rightarrow 1.$$

The exact sequence is said to split if H can be identified with a subgroup of G , so that $N \cap H = \{e\}$.

If N is normal, then observe that any subgroup H of G acts on N by conjugation; furthermore, conjugation determines a homomorphism

$$\iota: H \rightarrow \text{Aut}(N)$$

given by $h \mapsto \iota_h$, where $\iota_h(n) = hnh^{-1}$.

Now, we consider a certain tautology. If $G = NH$ with $N \cap H = \{e\}$ and N normal, then observe that

$$n_1 h_1 n_2 h_2 = (n_1 (h_1 n_2 h_1^{-1})) (h_1 h_2).$$

In particular, if we know the conjugation action of H on N , then we can recover the operation in G from this information and the operations in N and H .

This allows us to construct semidirect products by considering any two arbitrary groups N and H with an arbitrary group homomorphism

$$\begin{aligned} \theta: H &\rightarrow \text{Aut}(N) \\ h &\mapsto \theta_h. \end{aligned}$$

Then, we may define an operation on $N \times H$ by taking

$$(n_1, h_1) \cdot (n_2, h_2) = (n_1 \theta_{h_1}(n_2), h_1 h_2).$$

Observe that in the special case where $N, H \leq G$ with $G = NH$, N normal, and $N \cap H = \{e\}$, the operation θ can be given by the map ι . The group $(N \times H, \cdot)$ is denoted $N \rtimes_{\theta} H$, and is called the semidirect product of N and H with respect to θ . Observe that the ordinary direct product is a semidirect product where θ is the trivial map.

We will discuss some examples.

- (i) If G has a normal subgroup K of order n and a subgroup H of order m , with $HK = G$, $H \cap K = \{e\}$, and $\gcd(m, |\text{Aut}(K)|) = 1$, then $G \cong H \times K$.
- (ii) Suppose H is an abelian group of arbitrary order, and let $K = \mathbb{Z}/2\mathbb{Z}$. Define $\varphi: K \rightarrow \text{Aut}(H)$ by inversion. Then, $G = H \rtimes K$ contains the subgroup H of index 2, with $xhx^{-1} = h^{-1}$ for all $h \in H$ and $e \neq x \in K$.

When $H \cong \mathbb{Z}/n\mathbb{Z}$, this yields the dihedral group D_n of order $2n$, while if $H \cong \mathbb{Z}$, we have the infinite dihedral group D_{∞} .

- (iii) If H is a group, and $K = \text{Aut}(H)$, then $\varphi: K \rightarrow \text{Aut}(H)$ given by the identity yields the semidirect product $H \rtimes \text{Aut}(H)$, denoted the *holomorph* of H .

Solvable Groups

Definition: A series of subgroups G_i of a group G is a decreasing sequence of subgroups

$$G = G_0 \supset G_1 \supset G_2 \supset \cdots,$$

where the length of the series is the number of strict inclusions.

A normal series is a series of subgroups where $G_{i+1} \trianglelefteq G_i$ for each i . The number $\ell(G)$ is the length of a maximal normal series.

A *composition series* is a normal series

$$G = G_0 \supset G_1 \supset \cdots \supset G_n = \{e\}$$

where the successive quotients G_i/G_{i+1} are simple.

Theorem (Jordan–Hölder Theorem): Let G be a group, and suppose there are two composition series

$$\begin{aligned} G &= G_0 \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_n = \{e\} \\ G &= G'_0 \triangleright G'_1 \triangleright G'_2 \triangleright \cdots \triangleright G'_m = \{e\}. \end{aligned}$$

Then, $m = n$, and the quotient groups $H_i = G_i/G_{i+1}$ and $H'_i = G'_i/G'_{i+1}$ agree up to isomorphism and permutation.

Proof. Considering the following composition series

$$G = G_0 \triangleright G_1 \triangleright G_2 \triangleright \cdots \triangleright G_n = \{e\}$$

we induct on n . If $n = 0$, then G is trivial and there is nothing to prove.

Suppose $n > 0$, and consider another composition series

$$G = G'_0 \triangleright G'_1 \triangleright G'_2 \triangleright \cdots \triangleright G'_m = \{e\}.$$

If $G_1 = G'_1$, then the result follows from the induction hypothesis since G_1 has a composition series of length $n - 1$.

Therefore, we may assume that $G_1 \neq G'_1$. We observe that $G_1 G'_1 = G$, since $G_1 G'_1$ is normal in G and $G_1 \triangleright G_1 G'_1$, but there are no proper normal subgroups between G_1 and G since G/G_1 is simple.

Consider the group $K = G_1 \cap G'_1$. The distinct subgroups $G_i \cap K$ determine a composition series

$$K \triangleright K_1 \triangleright K_2 \triangleright \cdots \triangleright K_r = \{e\}.$$

By the Second Isomorphism Theorem, we have

$$\begin{aligned} \frac{G_1}{K} &= \frac{G_1}{G_1 \cap G'_1} \\ &\cong \frac{G_1 G'_1}{G'_1} \\ &= \frac{G}{G'_1} \end{aligned}$$

and $G'_1/K \cong G/G_1$. These quotients are simple, so we obtain two new composition series

$$\begin{aligned} G &\triangleright G_1 \triangleright K \triangleright K_1 \triangleright \cdots \triangleright \{e\} \\ G &\triangleright G'_1 \triangleright K \triangleright K_1 \triangleright \cdots \triangleright \{e\}. \end{aligned}$$

These composition series only differ at the first step. The two series have the same length and the same quotients. Next, we claim that the first of these series has the same length and quotients of the original series. Since

$$G_1 \triangleright K \triangleright K_1 \triangleright K_2 \triangleright \cdots \triangleright K_r = \{e\}$$

is a composition series for G_1 , the induction hypothesis gives that it must have the same length and quotients as the composition series

$$G_1 \triangleright G_2 \triangleright \cdots \triangleright G_n = \{e\}.$$

Therefore, applying the induction hypothesis, it follows that the series

$$G'_1 \triangleright K \triangleright K_1 \triangleright K_2 \triangleright \cdots \triangleright K_r = \{e\}$$

has the same length and quotients as

$$G'_1 \triangleright G'_2 \triangleright \cdots \triangleright G'_m = \{e\}.$$

□

Proposition: Let G be a group, and let N be a normal subgroup of G . Then, G has a composition series if and only if N and G/N have composition series. Furthermore,

$$\ell(G) = \ell(N) + \ell(G/N),$$

and the composition factors of G consist of the composition factors of N and of G/N .

Proof. If G/N has a composition series, then the subgroups appearing in it correspond to subgroups of G containing N with isomorphic quotients. Therefore, if both G/N and N have composition series, concatenating produces a composition series for G .

For the converse, suppose G has a composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\},$$

and that N is a normal subgroup of G . Taking intersections with N gives a sequence of subgroups

$$N = G \cap N \supsetneq G_1 \cap N \supsetneq \cdots \supsetneq \{e\} \cap N = \{e\}$$

such that $G_{i+1} \cap N$ is normal in $G_i \cap N$ for all i . We claim that this is a composition series upon deleting repetitions. For this, we will establish that the quotients $(G_i \cap N)/(G_{i+1} \cap N)$ are either trivial or isomorphic to G_i/G_{i+1} . Consider the homomorphism

$$G_i \cap N \hookrightarrow G_i \rightarrow \frac{G_i}{G_{i+1}}.$$

The kernel is $G_{i+1} \cap N$, so by the first isomorphism theorem, we have an injective homomorphism

$$\frac{G_i \cap N}{G_{i+1} \cap N} \hookrightarrow \frac{G_i}{G_{i+1}},$$

where we identify the image with a subgroup of G_i/G_{i+1} . This subgroup is normal since N is normal in G , and since G_i/G_{i+1} is simple, we get our desired result.

As for G/N , we have the composition series

$$\frac{G}{N} \supsetneq \frac{G_1 N}{N} \supsetneq \frac{G_2 N}{N} \supsetneq \cdots \supsetneq \{e_{G/N}\}.$$

such that $(G_{i+1} N)/N$ is normal in $(G_i N)/N$. By the third isomorphism theorem, the quotients

$$\frac{(G_i N)/N}{(G_{i+1} N)/N} \cong \frac{G_i N}{G_{i+1} N}$$

are isomorphic. Considering the homomorphism

$$G_i \hookrightarrow G_i N \rightarrow \frac{G_i N}{G_{i+1} N},$$

we have that this homomorphism is surjective, and the subgroup G_{i+1} of the source is sent to the identity of the target. By the first isomorphism theorem, there is a surjective homomorphism

$$\frac{G_i}{G_{i+1}} \rightarrow \frac{G_i N}{G_{i+1} N},$$

so that $G_i N/G_{i+1} N$ is either trivial or isomorphic to it as G_i/G_{i+1} is simple. $\square$

Composition series are the first major step in understanding the structure of groups. Next, we consider a special series known as the *derived series*.